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1 A

PDA.

The full form of PPA is push down automata, It is an automata which accepts context free language.

$$M = \{Q, \Sigma, q_0, F, Z_0, \delta, \#T\}$$

It has 3 components.

- Input Tape
- Automata.
- Stack.

B

FA

PDA.

- | | |
|--------------------|-------------------------|
| • Finite Automata | • Push down Automata. |
| • No stack memory | • Stack memory present. |
| • Regular language | • Context free language |

C

$$L = \{0^n 1^n \mid n \geq 0\}$$

$UVWXY$

Take pumping length = 4

$S = 00001111$

$U = 00$
 $V = 00$
 $W = 01$
 $X = 1$
 $Y = 11$

$|VX| \geq 1$
 $|VWX| \leq p$

UV^iWX^iY

pump $i = 4$

00 0000 01 1111 11

↳ belongs to language

So it is Context Free language

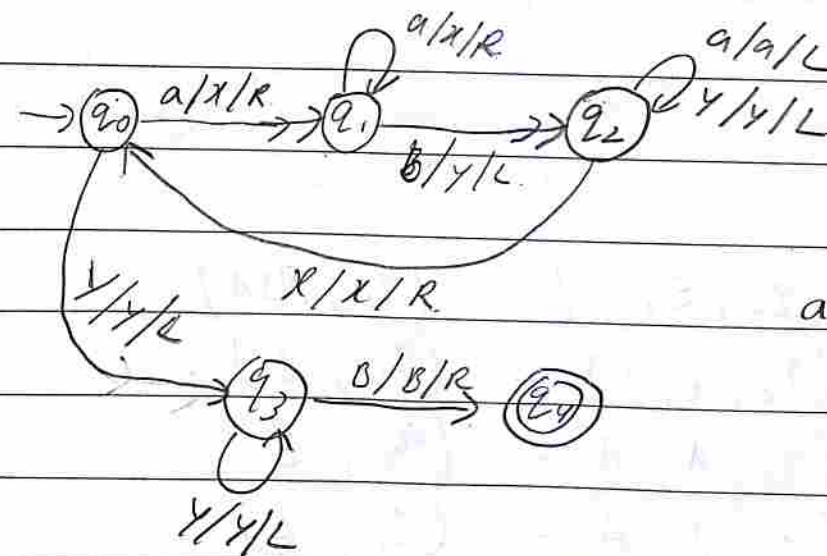
D

Turing Machine.

input / replace / direction.

A Turing machine accepts Recursive Enumerable Language.

$B a a b b B$



$a^n b^n \mid n \geq 0$

E

Subroutine in TM, it is when there is a routine in Turing's which is repeated. please I don't know.

F] HALTING → it is if Yes is given then Loop on else Stop completely.

2 A]

CFG $\rightarrow$ PDA

- ① $S \rightarrow ASA$
 - ② $S \rightarrow \epsilon$
 - ③ $A \rightarrow aB$
 - ④ $A \rightarrow d$
 - ⑤ $B \rightarrow C$
- $A \rightarrow a\gamma$
 $(q_0, a, A) \rightarrow (q_1, \gamma)$

- ① $(q_0, \epsilon, S) \rightarrow (q_1, ASA)$
- ② $(q_0, \epsilon, S) \rightarrow (q_1, \epsilon)$
- ③ $(q_0, a, A) \rightarrow (q_1, B)$
- ④ $(q_0, d, A) \rightarrow (q_1, \epsilon)$
- ⑤ $(q_0, C, B) \rightarrow (q_1, \epsilon)$

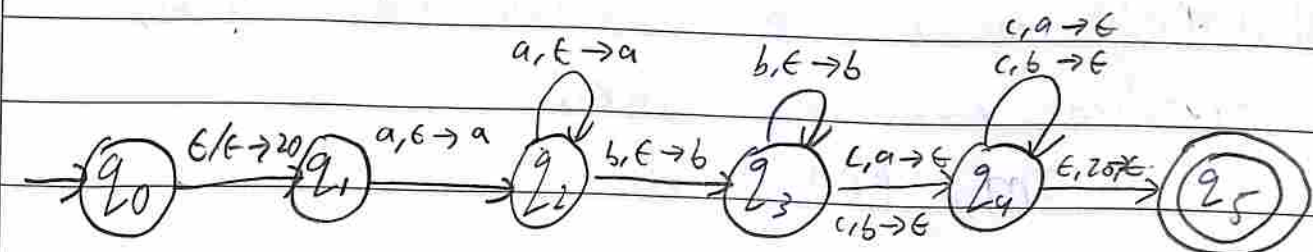
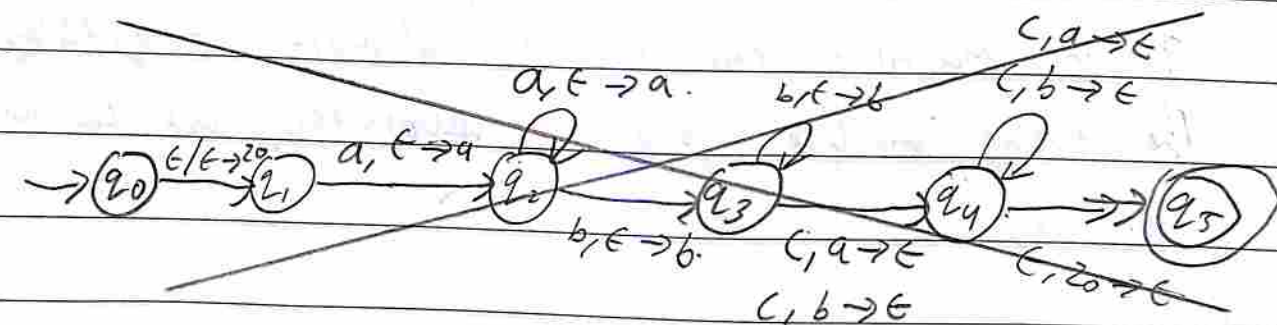
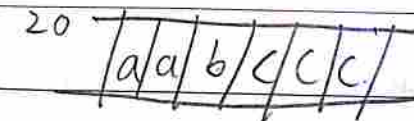
A PDA is constructed for the given CFG

B]

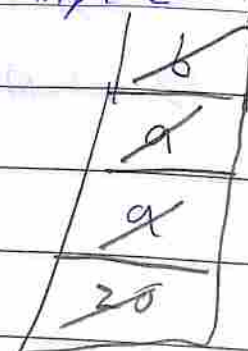
PDA.

input / pop $\rightarrow$ push.

$$L = \{a^n b^m c^{m+n} \mid m, n > 0\}$$



In this PDA it pushes a when input a, pushes b when input b and pops ^{both} a or b when input c.



D

Universal Turing Machine

A universal Turing machine is an automata which is accepted universally.

Turing machine can accept almost everything.

The Turing machine accepts everything and has no limitations to it.

A Turing machine is an automata which is using recursive enumerable language.

$TM > PDA > FA$

C

Chomsky Hierarchy is a hierarchy of the Chomsky.

The Chomsky Hierarchy is used when the Hierarchy of Chomsky is needed. There is also Chomsky normal form.

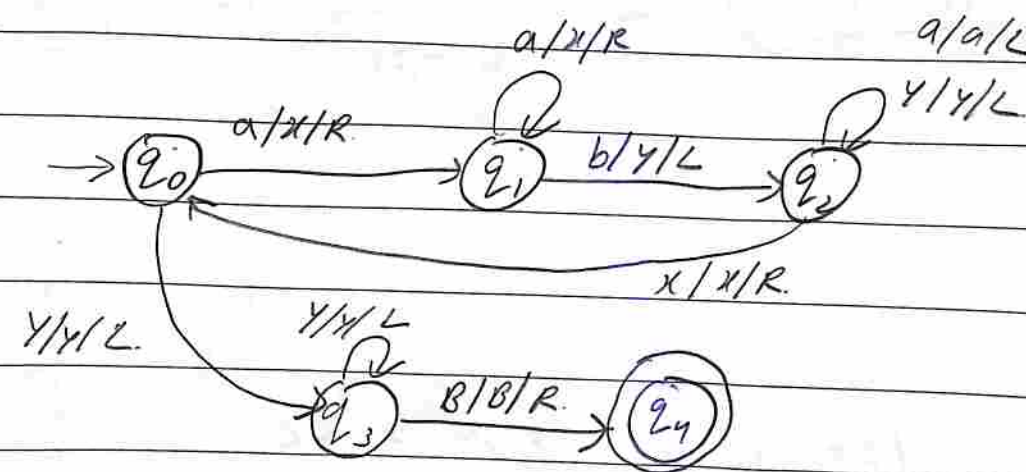
4 A

Linear Bounded Automata

Linear Bounded Automata uses context sensitive language

$$L = \{a^n b^n / n > 0\}$$

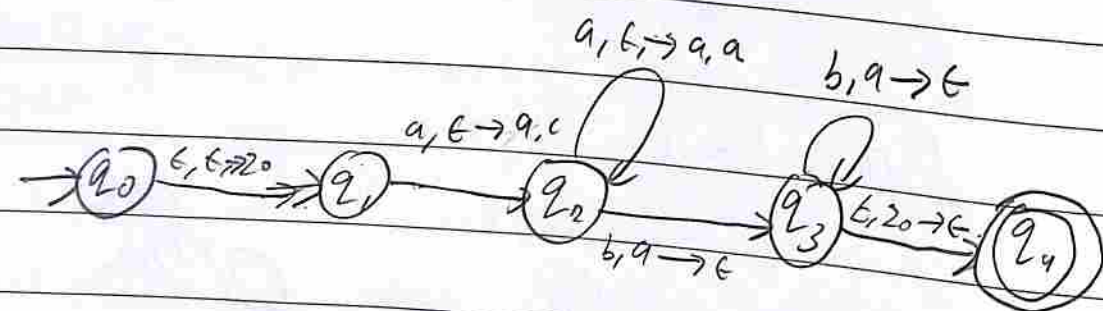
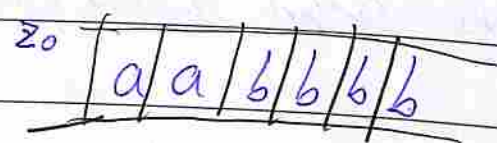
[a a b b]



B/

PDA

$$L = \{a^n b^{2n} \mid n \geq 0\} \quad w = \{a, b\}$$



In this PDA for every input a we push 2 a 's into the stack.
So for every input b we pop 1 a .

a
a
a
a
z_0

5 A/

Types of Turing machine

- Multi dimensional TM
- Non Deterministic TM.
- Two Way TM.
- Multi track TM.

B/

TM.

$$L = \{a^n b^n c^n \mid n \geq 0\}$$

aa b b c c

